

ITEM	WHAT IT WOULD DO	WHY IT WAITS, AND WHAT IT COSTS
01 for durability	A durable queue between collector and store ▪ Three brokers, on the three storage machines ▪ KRaft: Kafka keeps its own cluster state, with no separate coordination service beside it ▪ It decouples a slow reader from a fast writer Fluent Bit writes fast, OpenSearch reads at its own pace ▪ A slow or dead store then stops costing records	NEXT ▪ Nothing needs it yet: two readers today, served by two collector outputs ▪ A third reader is the trigger ▪ Costs three brokers and a second system to run ▪ Memory these five machines do not have
02 More log types	▪ Today: DDS, stdout, InfoLogger, collector health ▪ Missing: the machine's system journal, and the ten per-device process logs that slide 3 names ▪ Then parse numbers out of the message body: timeframe processing time, frame size, tracks	▪ The archive we replay holds neither source, so this one needs a real machine ▪ The parsers need real message shapes to test ▪ Costs one input and one parser per machine
03 Anomaly detection on the log text	▪ Today we watch numbers about the logs, not the text ▪ Slide 13 gives that gap a full page, with the three routes we are choosing between	▪ The choice between the routes is not made ▪ Nothing here is built
04 The shifter's own view	▪ The cockpit says whether the pipeline is healthy ▪ A shifter asks what the experiment is saying now ▪ A second view, close to the InfoLogger one	▪ It needs access to the EPN machines: we must read the running interface first ▪ No data work; the fields are already mapped
05 Authentication inside the cluster	▪ One password guards the web page ▪ Behind it, no port asks who you are ▪ A firewall rule is the whole boundary	▪ Fine on five machines we own, not on the farm ▪ Costs a certificate on every machine, an account for every job, and authentication on every call